

BE Assignment - 9

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Batch: 05

Code -

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Assignment-009.py > ...
1  from tkinter import *
2  import time
3
4  # Simulated Smart Contract
5  class SimpleStorageContract:
6      def __init__(self):
7          self.value = 0
8          self.transactions = []
9
10     def store(self, new_value):
11         self.value = new_value
12         tx = {
13             "value": new_value,
14             "time": time.strftime("%Y-%m-%d %H:%M:%S", time.localtime())
15         }
16         self.transactions.append(tx)
17         return tx
18
19     def retrieve(self):
20         return self.value
21
22 # GUI Application
23 class SimpleStorageApp:
24     def __init__(self, root):
25
26         self.contract = SimpleStorageContract()
27         self.root = root
28         self.root.title("Simple Storage Contract Simulation")
29
30         # Input field
31         self.entry = Entry(root, width=20)
32         self.entry.pack(pady=10)
33
34         # Store button
35         self.store_button = Button(root, text="Store Value", command=self.store_value)
36         self.store_button.pack(pady=5)
37
38         # Status label
39         self.status_label = Label(root, text="")
40         self.status_label.pack(pady=5)
41
42         # Transaction hash label
43         self.hash_label = Label(root, text="")
44         self.hash_label.pack(pady=5)
45
46         # History label
47         self.history_label = Label(root, text="Transaction History:")
```

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47     self.history_label.pack(pady=10)
48
49     def store_value(self):
50         try:
51             value = int(self.entry.get())
52             tx = self.contract.store(value)
53             self.status_label.config(text="Value stored successfully!", fg="green")
54             self.hash_label.config(text=f"Stored Value: {tx['value']}\nTime: {tx['time']}")
55             self.update_history()
56         except ValueError:
57             self.status_label.config(text="Please enter a valid integer.", fg="red")
58
59     def update_history(self):
60         history_text = "Transaction History:\n"
61         for tx in self.contract.transactions:
62             history_text += f"Value: {tx['value']} | Time: {tx['time']}\n"
63         self.history_label.config(text=history_text)
64
65 if __name__ == "__main__":
66     root = Tk()
67     app = SimpleStorageApp(root)
68     root.mainloop()

```

Output -

